

Beyond vectors — structured knowledge

LESSON 01 · GROUND IN KNOWLEDGE — STUDIO: KNOWLEDGE GRAPH & SEMANTIC MODEL

What you'll learn

- Classify enterprise data into three structural shapes: text (unstructured), connections (relational graphs), and tables (structured SQL databases).
- Select the correct Lyzr retrieval architecture (Knowledge Base, Knowledge Graph, or Semantic Data Model) to match the shape of the data.
- Understand the limits of vector search (RAG) when querying interconnected policy rules or structured tabular metrics.
- Stack multiple knowledge architectures within a single agent to handle multi-modal enterprise queries.

Key concepts

Knowledge comes in three distinct shapes, each requiring a different retrieval architecture. The three shapes are: paragraphs of text (solved by basic Knowledge Bases), connected entities and relationships (solved by Knowledge Graphs), and tabular data (solved by Semantic Data Models).

Basic vector RAG is the wrong tool for questions that require exact relational hops. Standard vector retrieval chunks documents and searches by similarity. It fails when a question relies on understanding how different policy conditions, exemptions, or entity nodes relate to each other.

Knowledge Graphs store entities and relationships, mapping how facts connect. By representing information as nodes (entities) and edges (relationships), a Knowledge Graph allows agents to hop across connections (e.g., warranty conditions linked to product tiers) to synthesize complex logic.

Semantic Data Models teach agents database schemas to run exact computations. When querying numerical values like total sales or order counts, agents must write and execute SQL queries against relational tables instead of searching for "similar paragraphs" about revenue.

One agent can carry all three knowledge types, stacking their capabilities. A production agent does not need to choose a single architecture; it can retrieve unstructured context from a knowledge base, navigate connections in a graph, and query database tables in parallel.

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In Studio

Studio feature: **Build → Knowledge**. Here's a tour of the three tabs that represent the shapes of data:

1. Navigate to the **Knowledge** screen under **Build** in the left-hand navigation panel.
2. Note the three tabs at the top: **Knowledge Base**, **Knowledge Graph**, and **Semantic Data Model**.
3. Choose **Knowledge Base** when uploading manuals, policies, or markdown docs where passage similarity search is sufficient.
4. Choose **Knowledge Graph** to connect entity relationships extracted from text and synchronize them with a graph database like Neo4j.
5. Choose **Semantic Data Model** to teach the agent your relational database schema (SQL) for exact calculations and aggregations.
6. Under agent settings, attach one, two, or all three knowledge assets to a single agent to combine their retrieval paths.

TRY THIS

- 1 Match these three customer service queries to their correct knowledge architecture:

- "What is our company's refund policy?" - "Which customer account manager handles the distributor that bought our model X?" - "How many refunds did we process yesterday?"

- 1 Open the **Knowledge** section in Studio. Navigate between the three tabs and inspect their different connection requirements.